

**2025 – Lab Exam 04
Report**

Student ID	
Batch	
Marking Guide	
1. DAO'S	3
2. ORM : Data classes for entities	3
3. CRUD Operations	4
Total Marks	10
Evaluator	

OneDrive Project Link:

[Habbittracker](#)

Description:

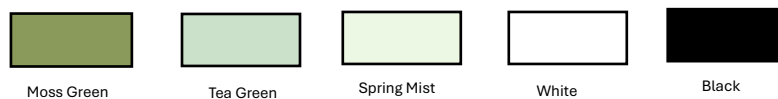
HappyMe is a personal wellness and habit-tracking Android app designed to motivate users toward a healthier and happier lifestyle. It helps users manage daily habits, monitor mood changes, and maintain proper hydration — all in one place. The app focuses on positivity and mindfulness through its calm design, motivational messages, and easy-to-use features.

Main Features

- **Home Page:** Displays daily to-do habits, water-drinking progress, motivational quotes, and a 7-day mood summary bar chart. When a habit is marked as done, it disappears from the list to maintain focus. Tapping a mood bar reveals the mood of that particular day.
- **Routine Page:** Allows users to manage habits - add, update, and view history. For first-time users, onboarding includes entering their name and selecting preferred habits. The user's name with a greeting ("Good Morning/Good Afternoon, (name)") appears on top, creating a personalized feel. Completed habits move to the bottom, while pending ones stay at the top. Users can also generate a PDF report of their progress.
- **Mind Page:** Focuses on emotional wellness. Users can select positive moods, add reflective notes under "*Want to reflect? Write here,*" and view their mood history. PDF summaries can also be generated for personal insights.
- **Fuel Page:** Encourages hydration as the "fuel of life." Users can add or remove cups of water, filling an 8-cup bottle for daily goals. Hydration reminders notify users until they meet their goal, then reset automatically at midnight.
- **MeTime Page:** Serves as a profile and settings page where users can view stats like total habits completed and day streaks. It also allows editing preferences, switching themes (including Night Mode), viewing history, and logging out.

Colors:

The color palette of "HappyMe" is based on natural green tones to create a motivational and peaceful experience:



- Moss green is used for buttons and selected parts of the navigation bar. It represents growth, balance, and renewal, aligning with the app's focus on self-improvement and consistency.
- Tea green is applied to highlights and progress indicators, symbolizing fresh starts and success.
- Spring Mist is used for the light mode background to give a soft and clean appearance, promoting relaxation and clarity.
- Black and white is used for text and icons to maintain clarity, simply, and elegance.

The green-based design was chosen because green is psychologically associated with calmness, healing, and motivation. It subconsciously encourages users to keep improving without stress. The overall palette reflects a natural, refreshing, and positive atmosphere, which supports the app's purpose of promoting happiness and wellness.

Technical Implementation

This application is built on the MVVM with Repository architecture, ensuring a clean separation of concerns where Fragments manage the UI, ViewModels handle the data, and Repositories abstract all data sources. For robust data management, it utilizes the Room Persistence Library to store structured user data—such as profiles, habits, mood entries, and hydration history—with all operations performed on background threads using coroutines to maintain smooth performance. The implementation enforces multi-user data isolation through `user_id` filtering in every query, while Kotlin Flow enables real-time UI updates for a reactive experience. The data model leverages entity relationships and foreign keys for consistency, and `SharedPreferences` is used alongside Room to efficiently cache simple configuration data, creating a optimized and responsive user application.

CRUD Operations

CREATE:

@Insert operations for new users, habits, completions, mood entries, and hydration records. All data creation happens on background threads.

READ:

@Query methods with Flow for real-time data streaming.

Includes `getSelectedHabits()`, `getTodayCompletions()`, and history queries, all user-filtered.

UPDATE:

@Update for modifying habits, completion status, hydration levels, and reminder settings. Handles daily progress tracking and user preferences.

DELETE:

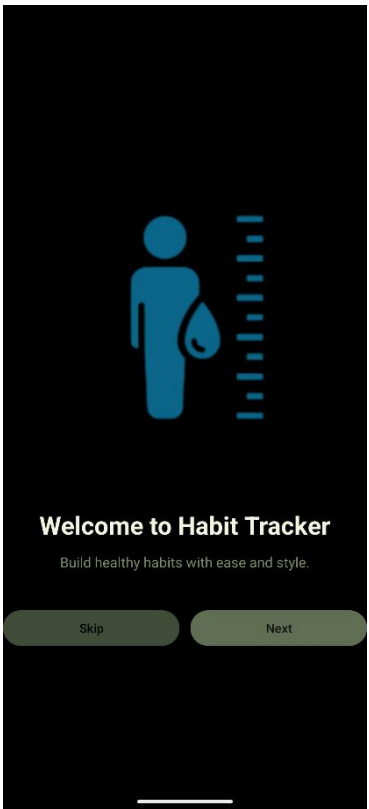
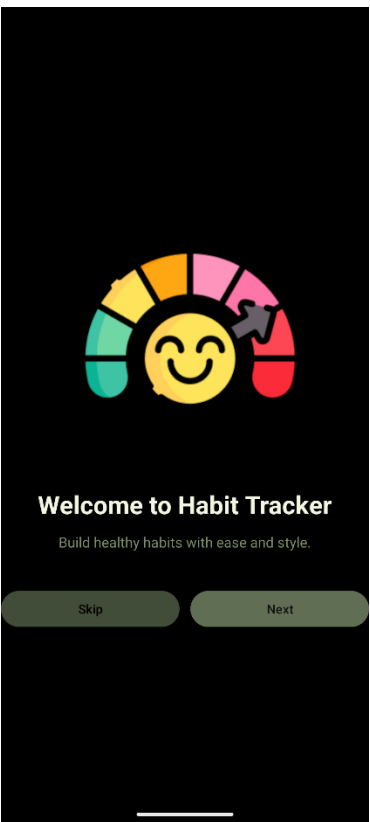
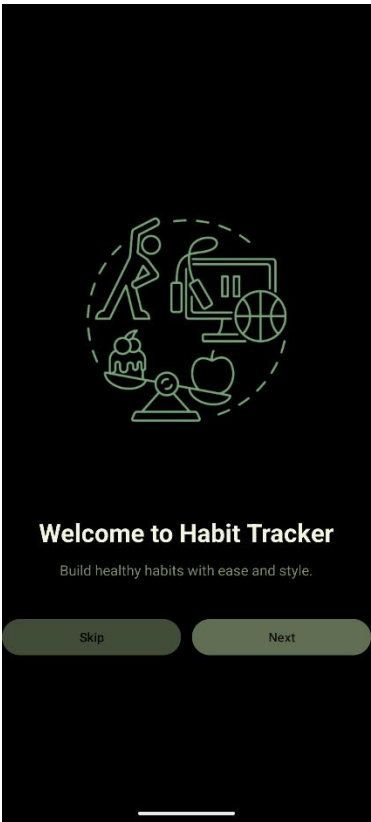
@Delete for data cleanup - daily resets clear completions, logout clears user data. Maintains database performance and data lifecycle.

Smart Operations:

Combined CRUD with business logic. `toggleHabitCompletion()` either inserts or updates based on existing records. Transaction safety for critical operations like daily resets.

The name **“HappyMe”** represents the idea of becoming a better and happier version of oneself through small, consistent actions each day. It encourages users to focus on self-improvement, positivity, and personal wellness rather than perfection. The app's motivational experience is built around this concept — every completed habit, filled water bottle, and mood reflection brings a sense of progress and calm. Combined with its soothing green color theme and uplifting quotes, *HappyMe* creates a refreshing and encouraging environment that helps users stay motivated, balanced, and emotionally uplifted throughout their wellness journey.

Screenshots:



4:13   

Pick your daily habits:



Wake up early
Start your day with energy!



Plan your day
Set goals, achieve more!



Mindfulness
Be present, stay calm!



Move your body
Stay active, stay fit!



Eat healthy
Fuel your body right!



Practice gratitude
Appreciate the little things!



Take a digital break
Unplug to recharge!



Get enough sleep
Recharge your body and mind!

[Save Habits](#)

4:13   



Tell us your name

[Next](#)



Sign Up

[Sign Up](#)

☐ Remember Me

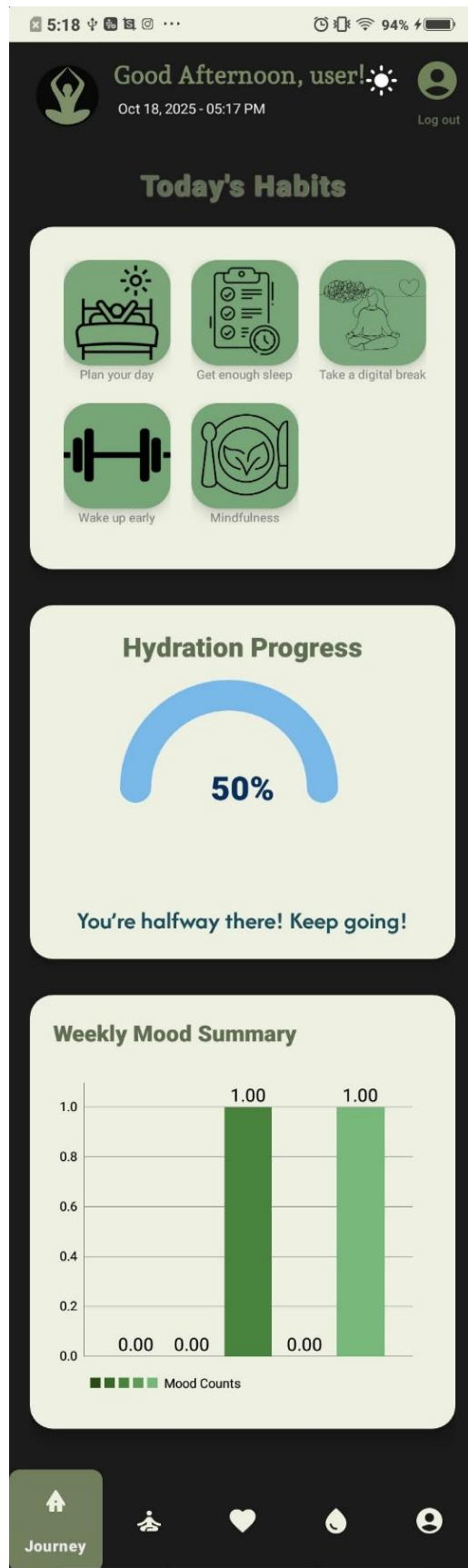
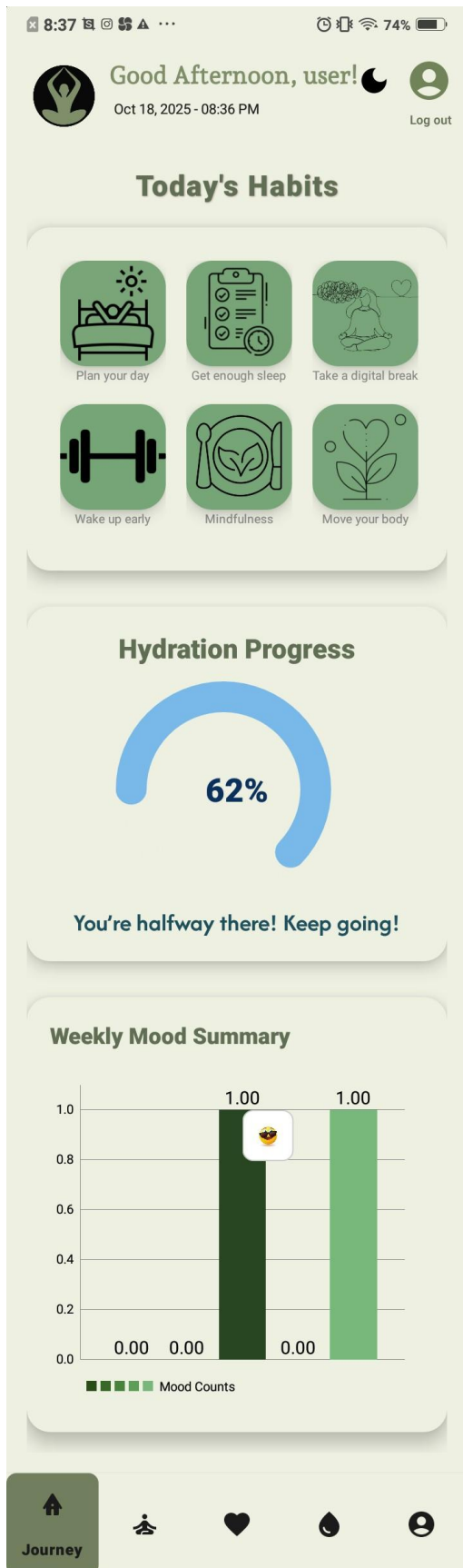
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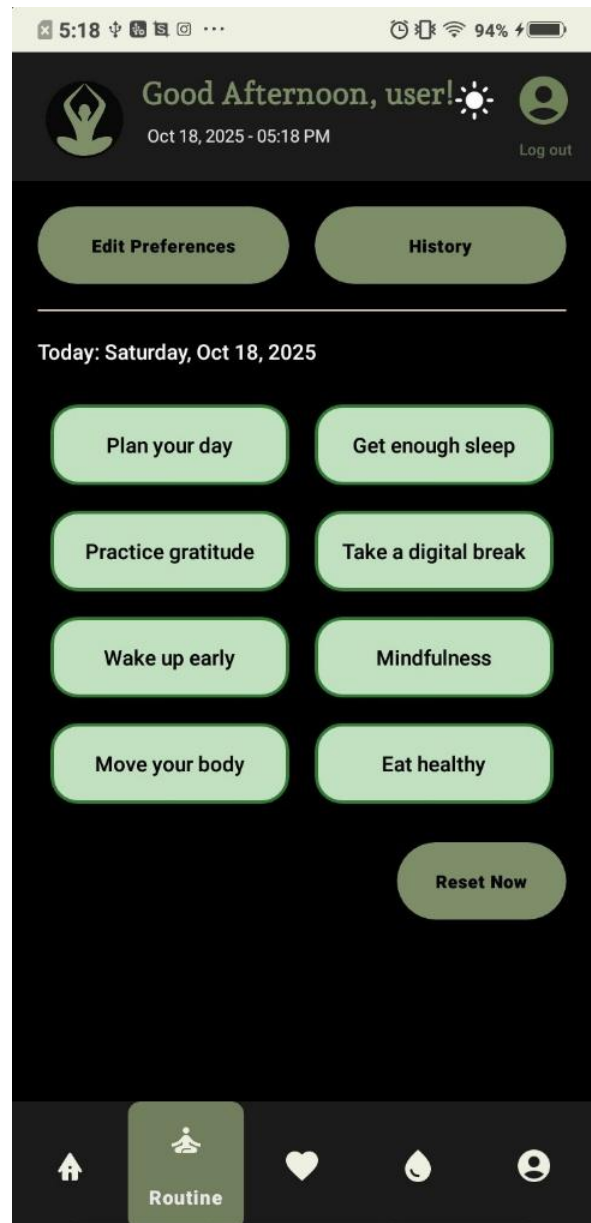
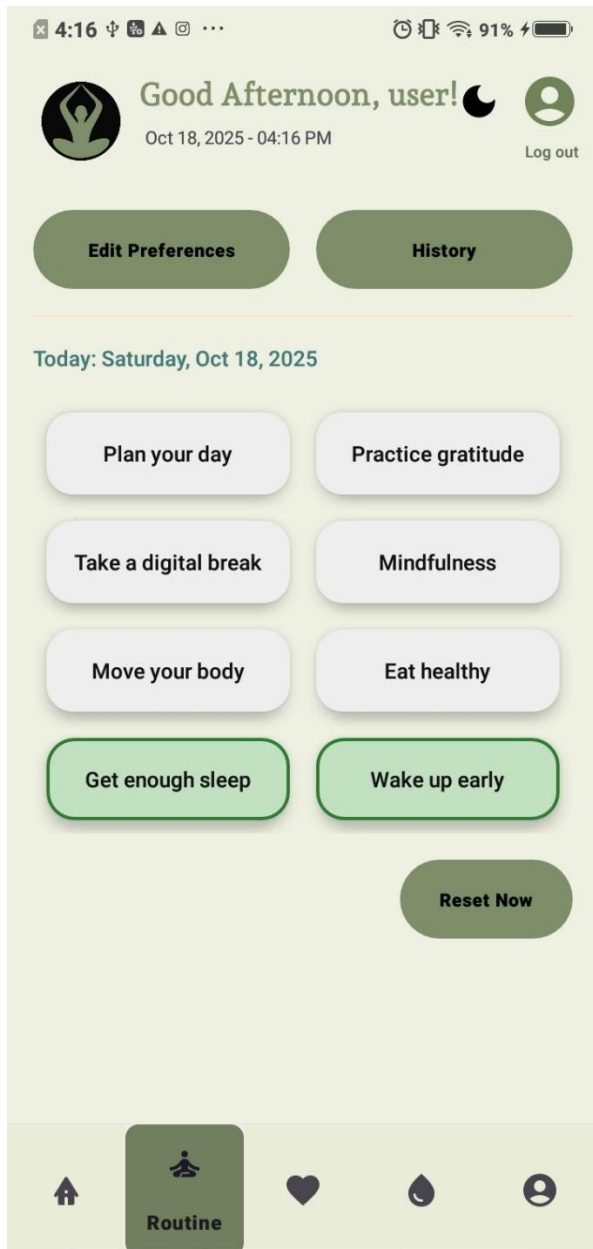


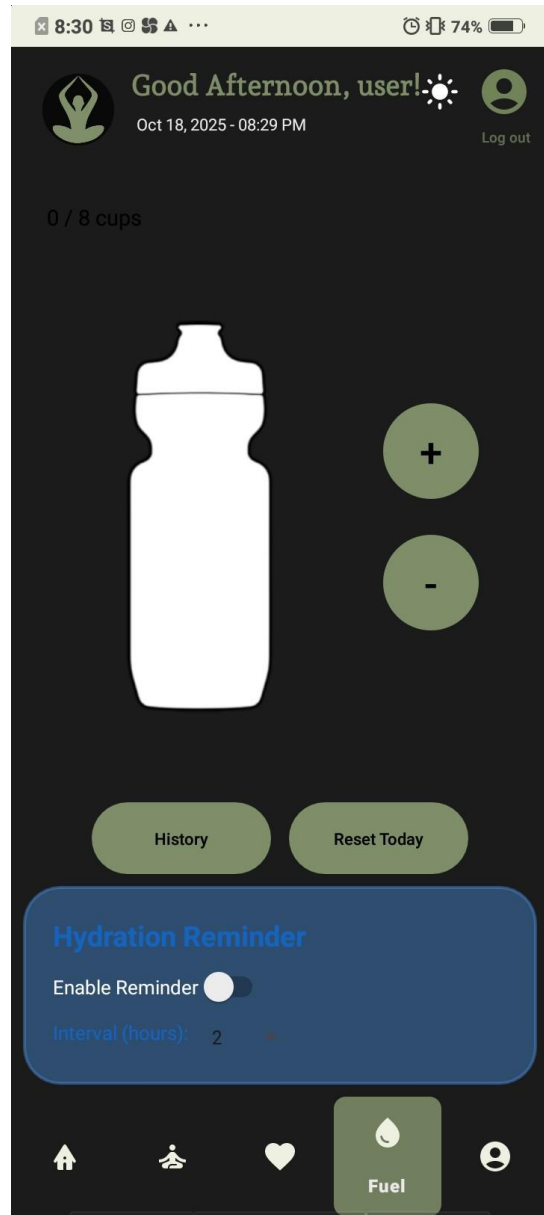
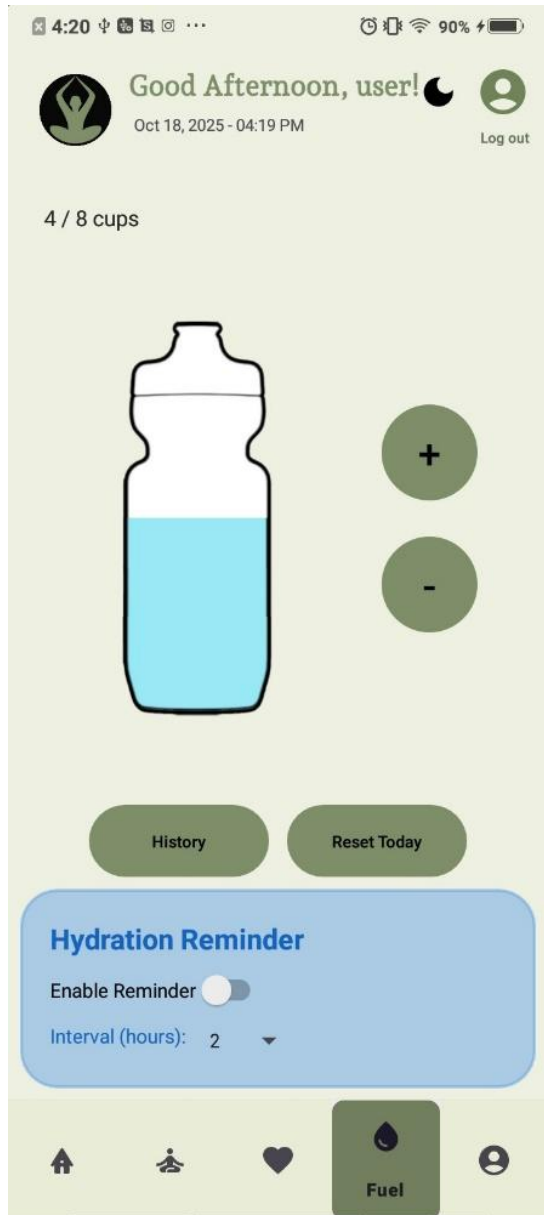
Login

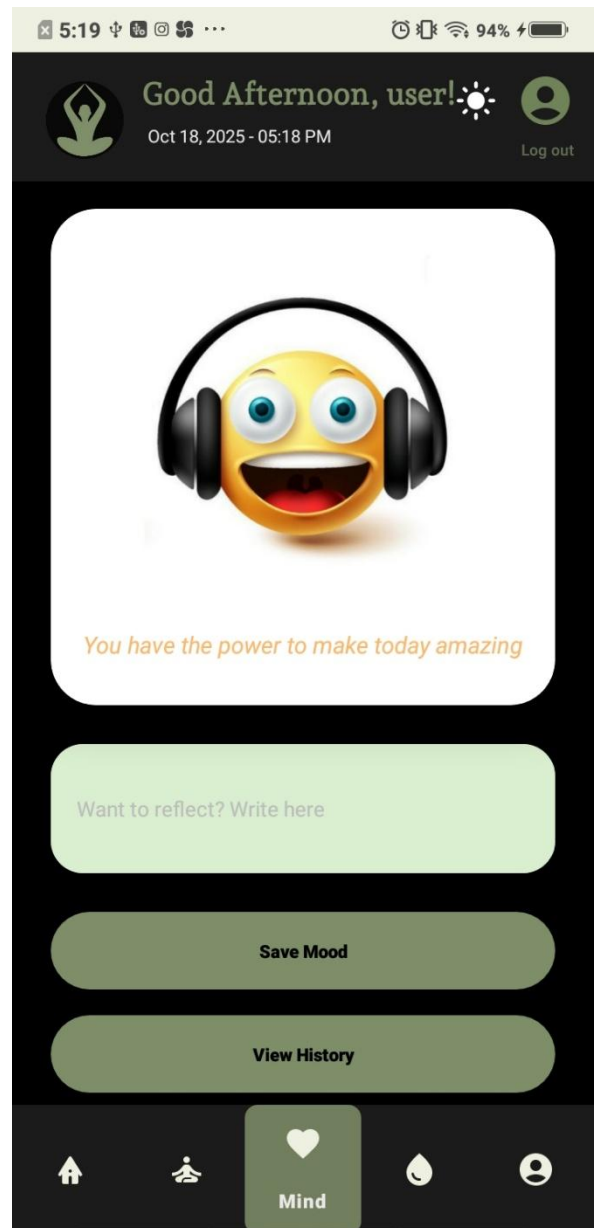
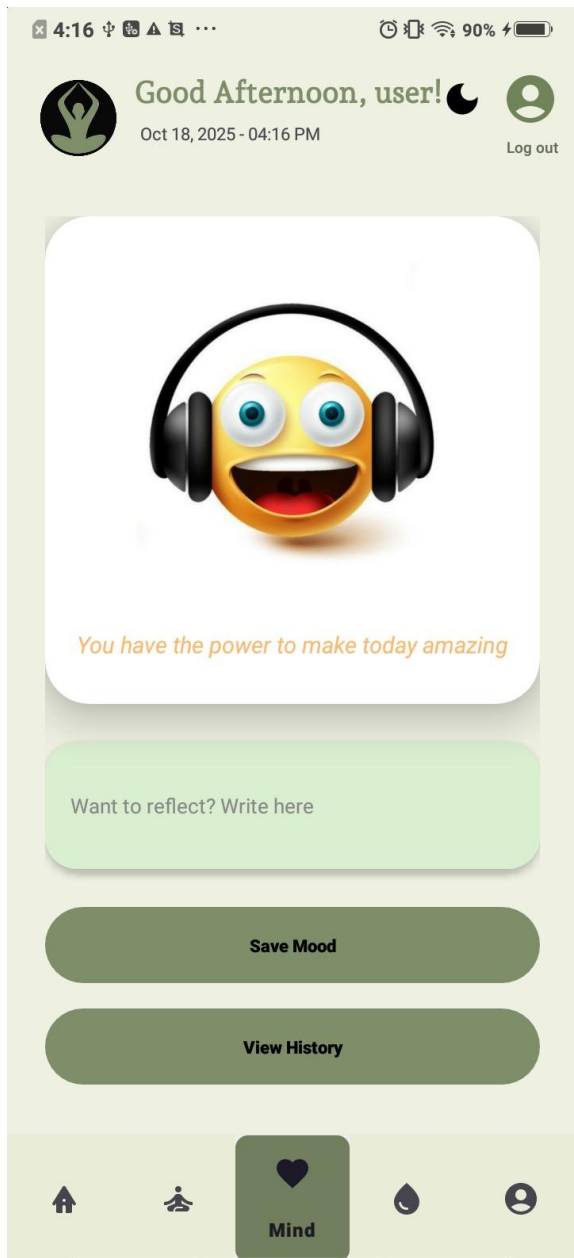
[Login](#)

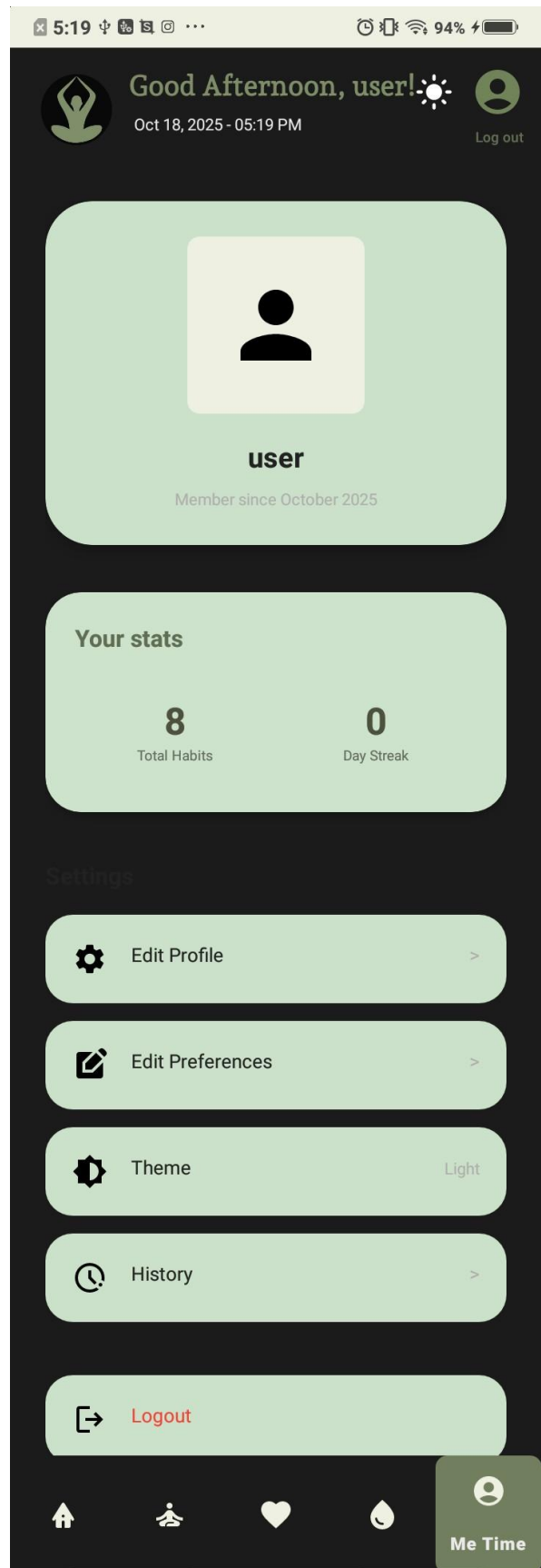
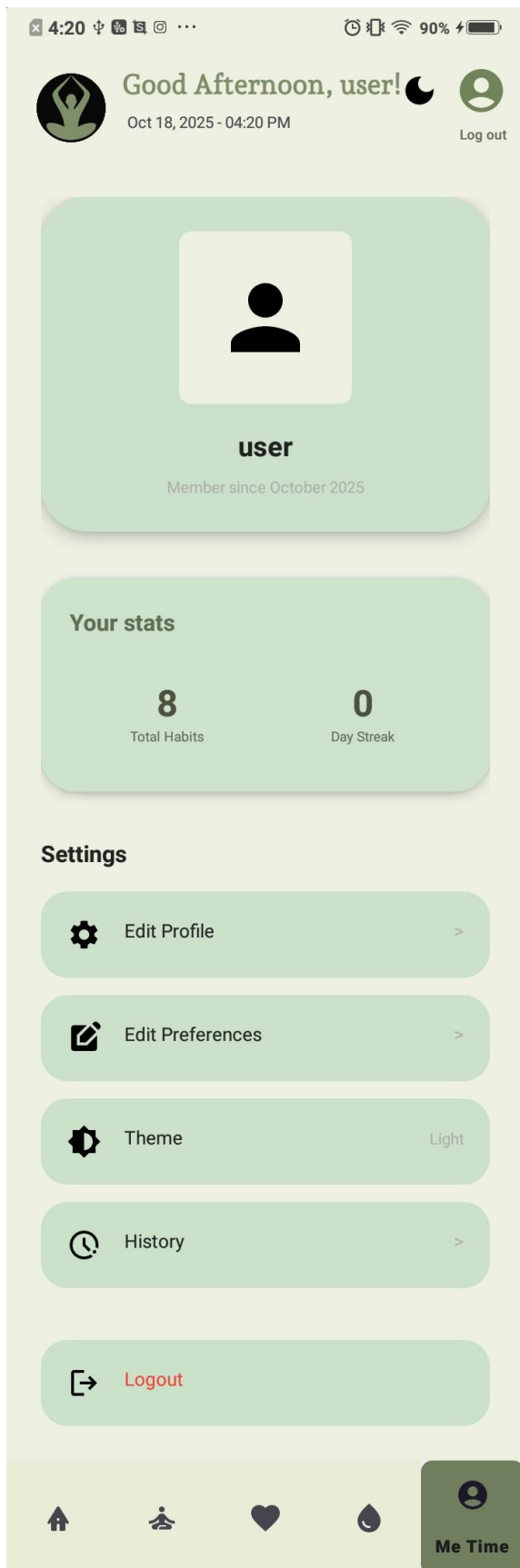
[Don't have an account? Sign up](#)

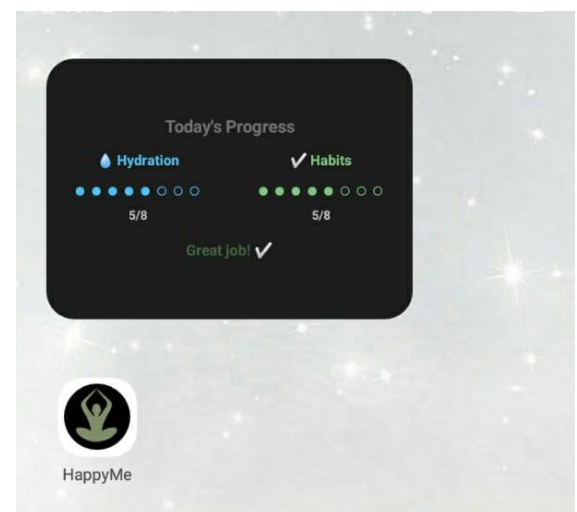
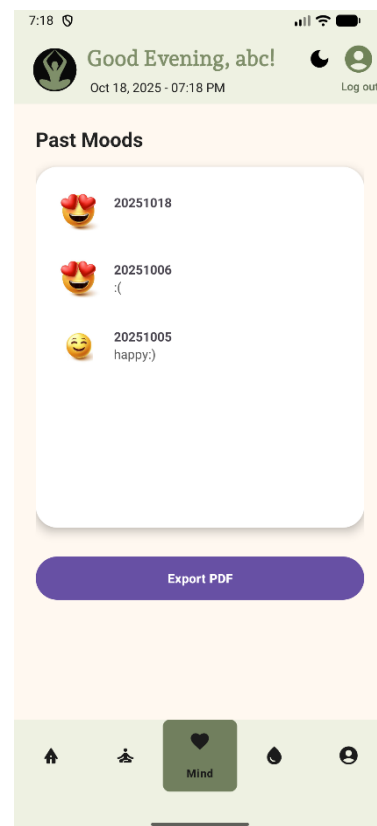
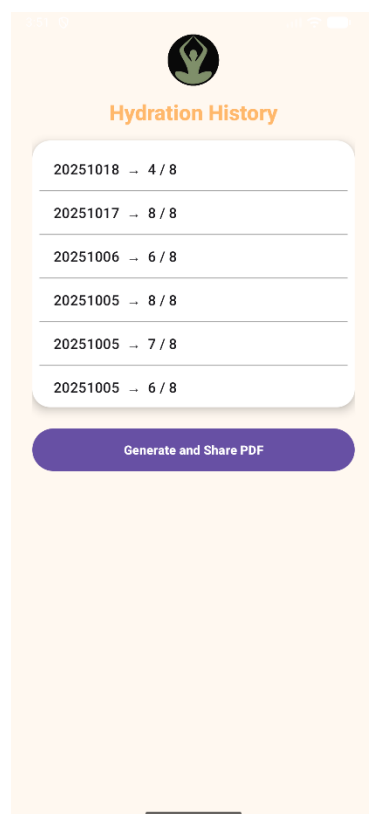
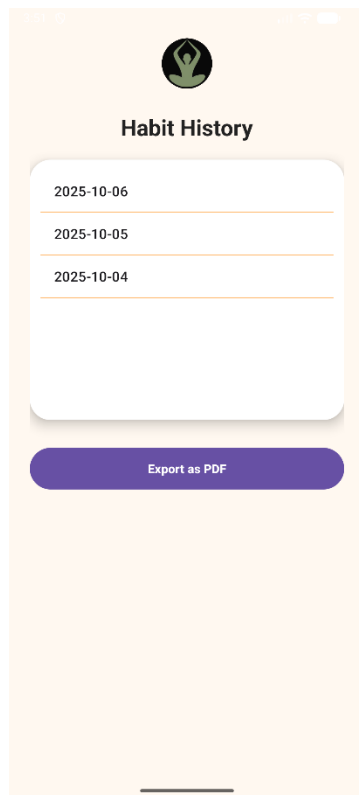






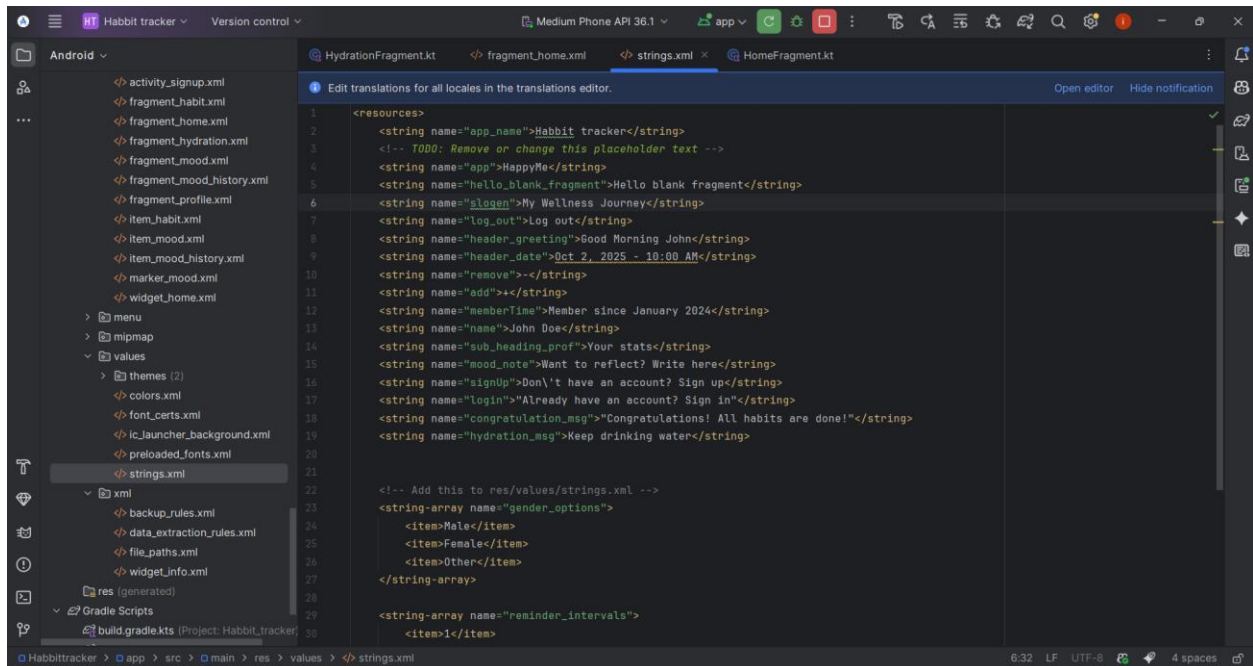




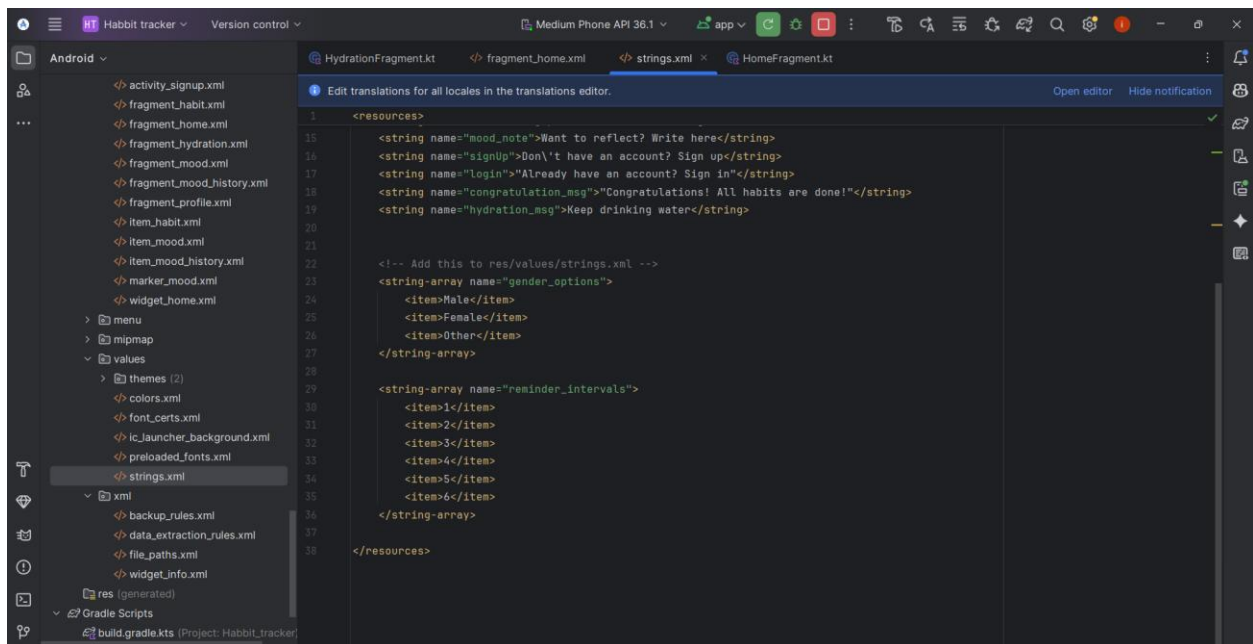


Content of xml files of Strings and Colors:

Strings



```
1 <resources>
2   <string name="app_name">Habit tracker</string>
3   <!-- TODO: Remove or change this placeholder text -->
4   <string name="app">HappyMe</string>
5   <string name="hello_blank_fragment">Hello blank fragment</string>
6   <string name="slogen">My Wellness Journey</string>
7   <string name="log_out">Log out</string>
8   <string name="header_greeting">Good Morning John</string>
9   <string name="header_date">Oct 2, 2025 - 10:00 AM</string>
10  <string name="remove"></string>
11  <string name="add">+</string>
12  <string name="memberTime">Member since January 2024</string>
13  <string name="name">John Doe</string>
14  <string name="sub_heading_prof">Your stats</string>
15  <string name="mood_note">Want to reflect? Write here</string>
16  <string name="signUp">Don't have an account? Sign up</string>
17  <string name="login">Already have an account? Sign in</string>
18  <string name="congratulation_msg">Congratulations! All habits are done!</string>
19  <string name="hydration_msg">Keep drinking water</string>
20
21
22  <!-- Add this to res/values/strings.xml -->
23  <string-array name="gender_options">
24    <item>Male</item>
25    <item>Female</item>
26    <item>Other</item>
27  </string-array>
28
29  <string-array name="reminder_intervals">
30    <item>1</item>
31  </string-array>
32 </resources>
```



```
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15  </string-array>
16
17  <string-array name="reminder_intervals">
18    <item>1</item>
19    <item>2</item>
20    <item>3</item>
21    <item>4</item>
22    <item>5</item>
23    <item>6</item>
24  </string-array>
25
26 </resources>
```

Colors

